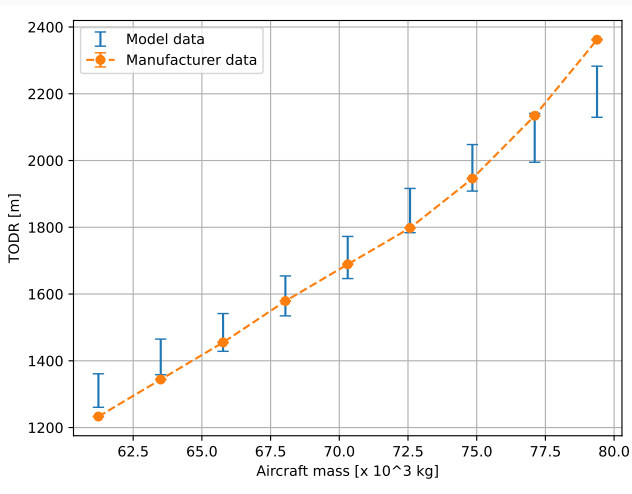


TODR, MTOM and lift coefficient

Lorenzo Cane

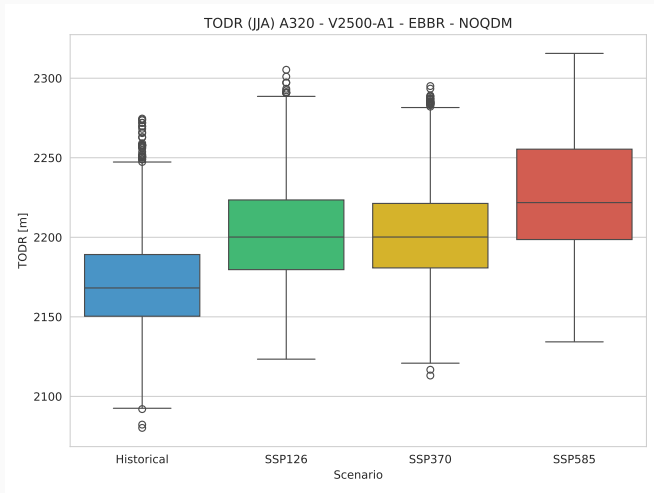
May 5, 2025

$$C_L = 1.61$$



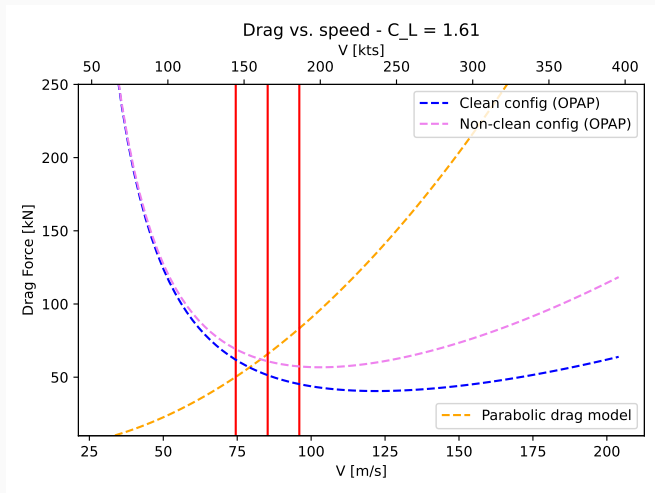
Using Drag, friction, $\Delta v 0.01$ m/s.

$$C_L = 1.61$$



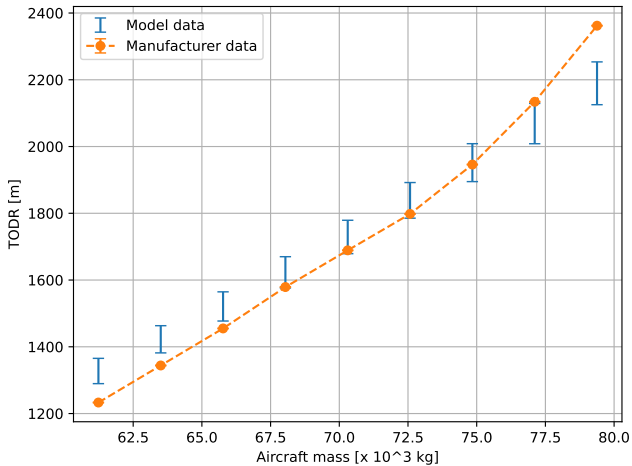
EBBR Airport TODR vs scenarios.

Drag from OpenAP



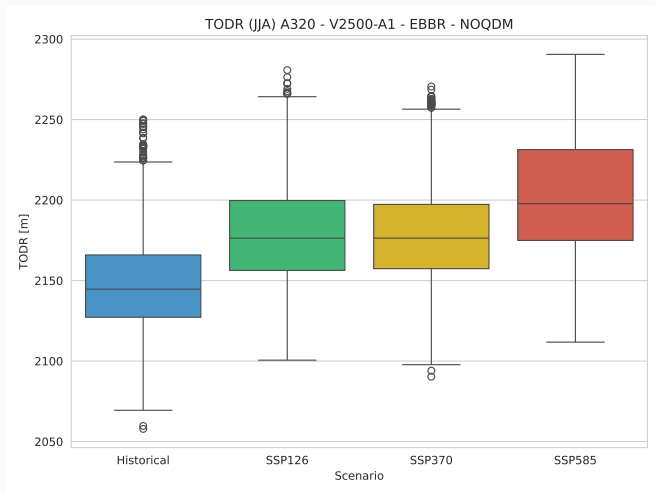
Drag from OpenAP is suitable for cruise phase.

Without drag (only gears friction)



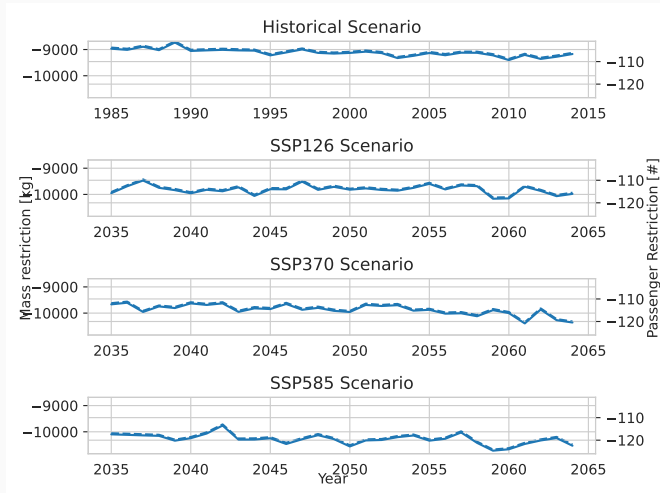
$$C_L = 1.35 \pm 0.04.$$

Without drag (only gears friction)



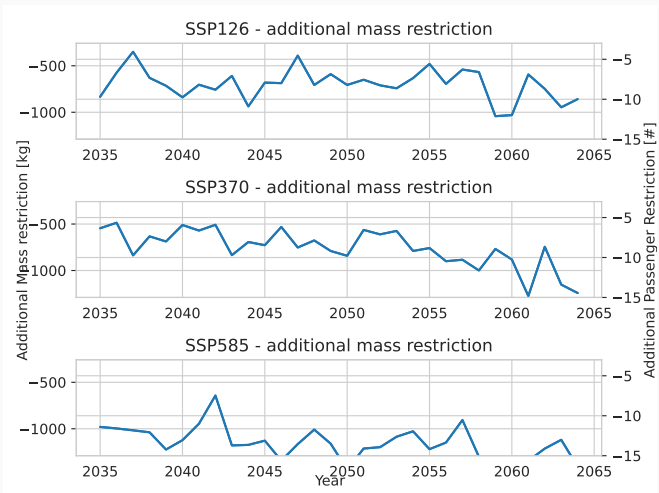
TODR is underestimated.

Mass restriction



MTOM for LIGC airport using $C_L = 1.61$. Passenger mass: $m_p = 86.0$ kg.

Additional mass restriction



Additional mass restriction for LIGC airport using $C_L = 1.61$. Passenger mass: $m_p = 86.0$ kg. The average mass restriction from the historical period (1985-2014) has been subtracted from the scenarios mass restriction.